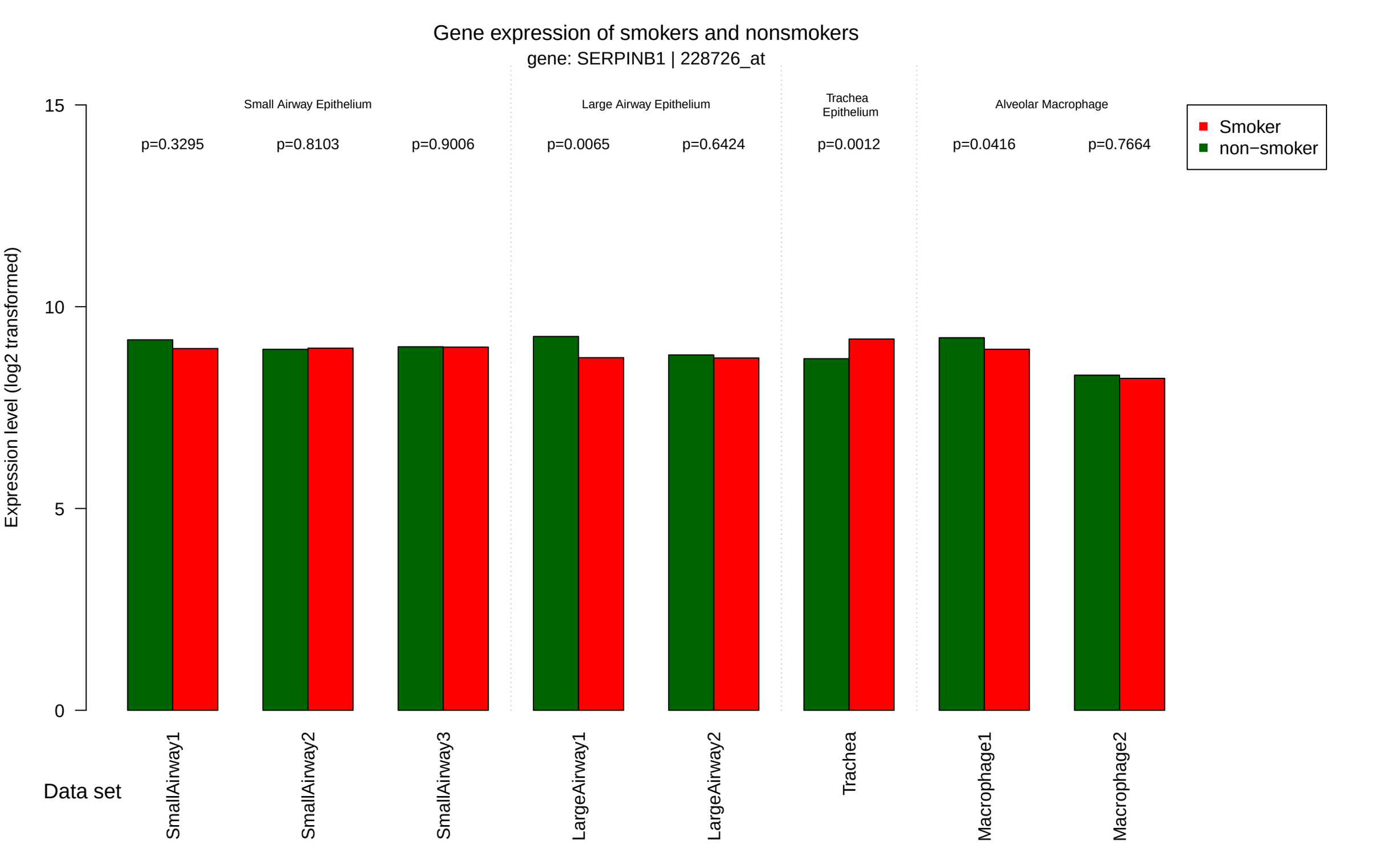
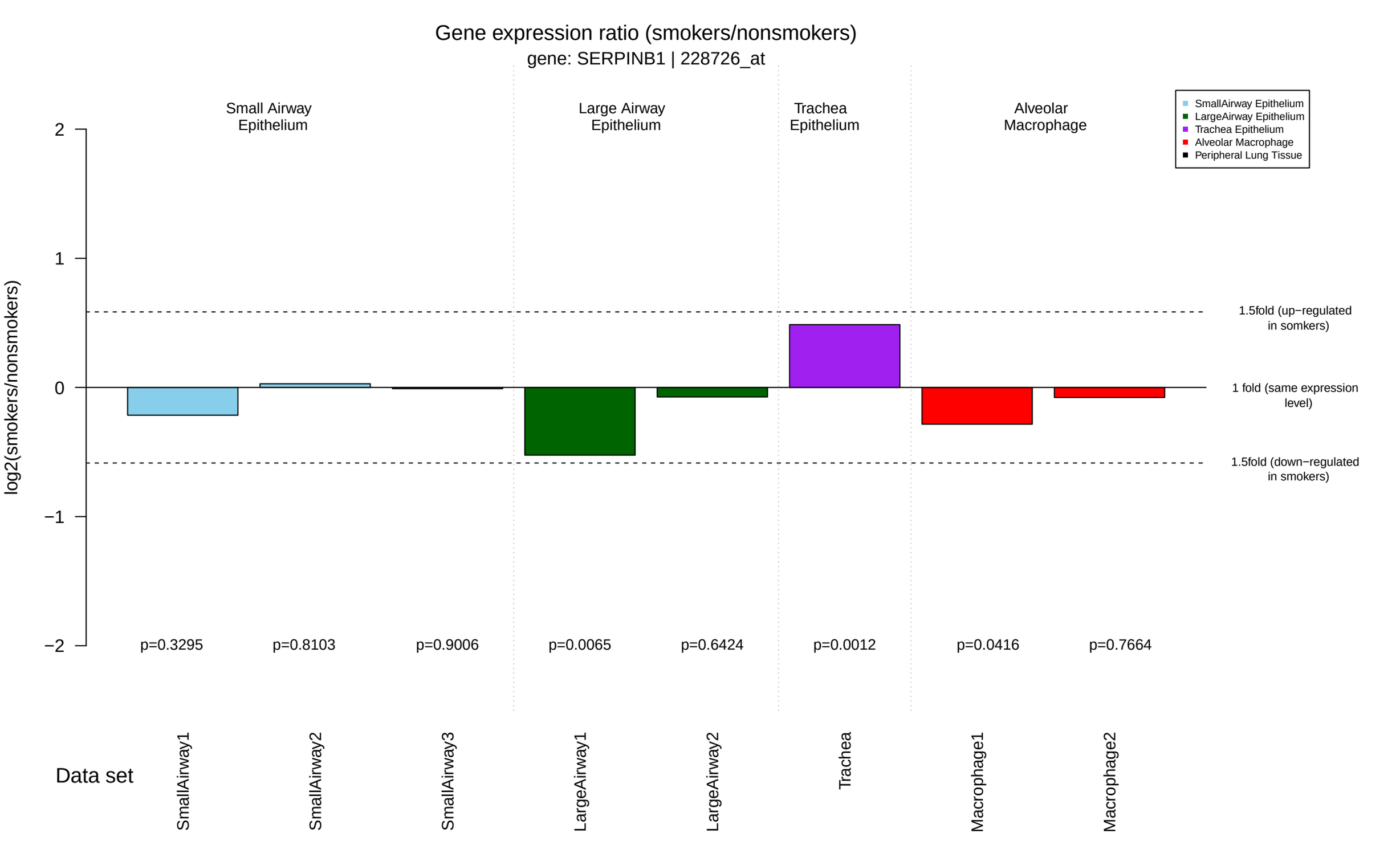
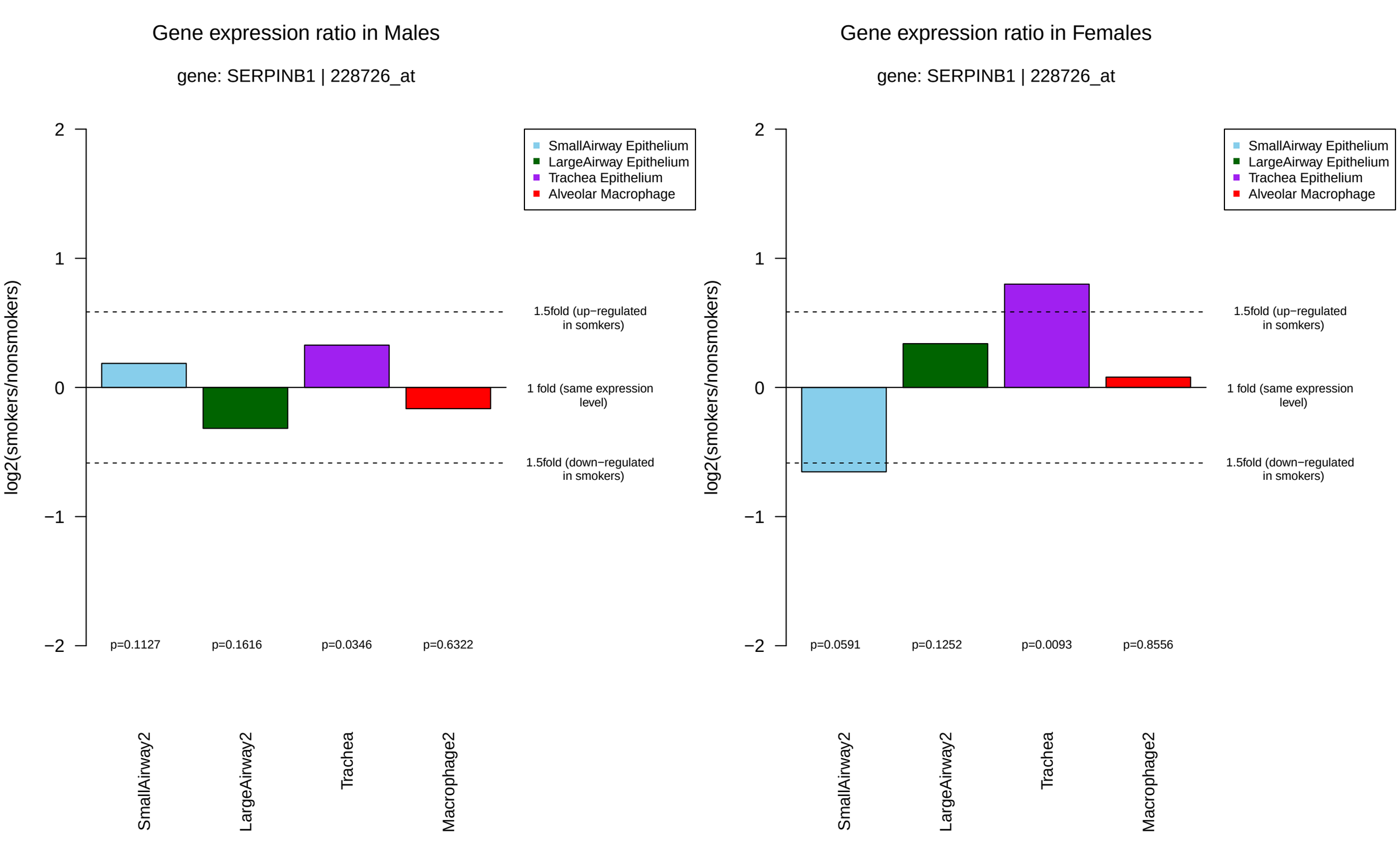
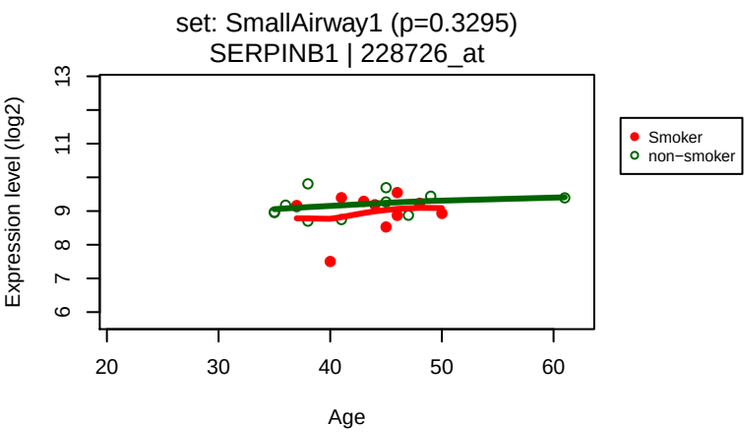
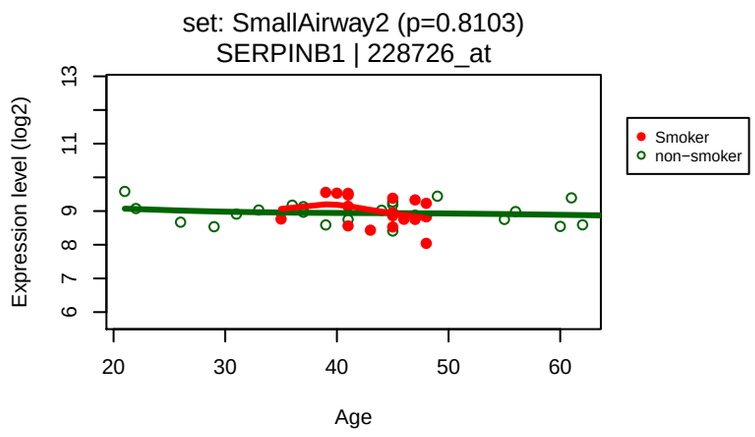
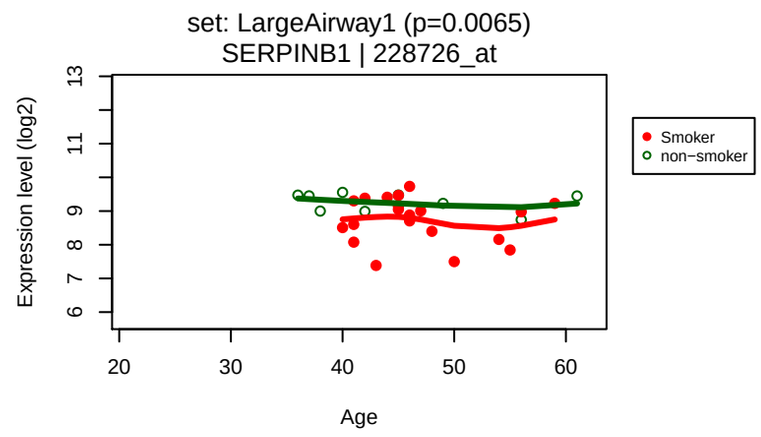

Gene expression of smokers and nonsmokers
gene: SERPINB1 | 228726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3295
p=0.8103
p=0.9006
p=0.0065
p=0.6424
p=0.0012
p=0.0416
p=0.7664
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

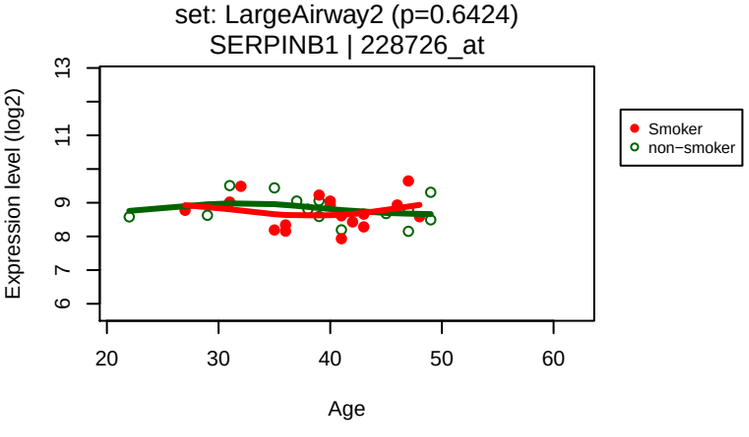

Gene expression ratio (smokers/nonsmokers)
gene: SERPINB1 | 228726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3295
p=0.8103
p=0.9006
p=0.0065
p=0.6424
p=0.0012
p=0.0416
p=0.7664
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

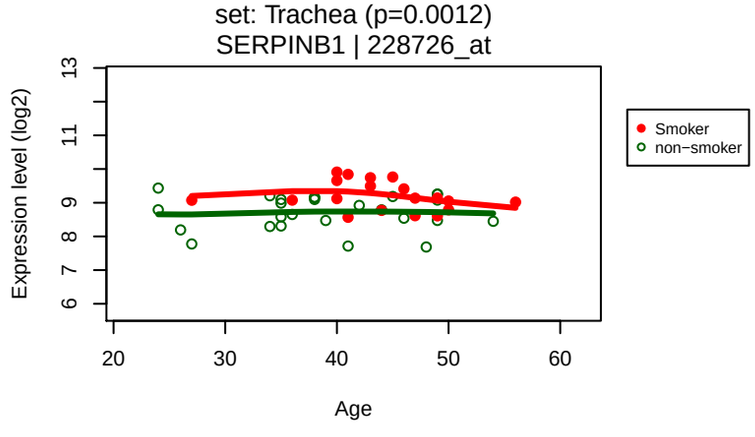

Gene expression ratio in Males
gene: SERPINB1 | 228726_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1127
p=0.1616
p=0.0346
p=0.6322
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERPINB1 | 228726_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0591
p=0.1252
p=0.0093
p=0.8556
SmallAirway2
LargeAirway2
Trachea
Macrophage2

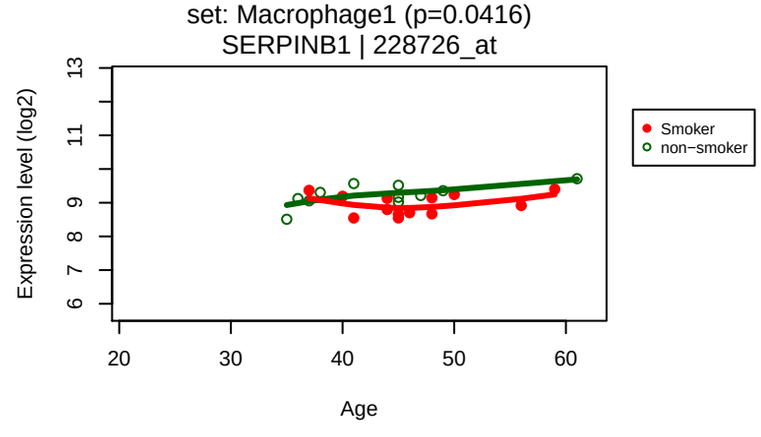

set: SmallAirway1 (p=0.3295)
SERPINB1 | 228726_at
Expression level (log2)
Age
Smoker
non−smoker

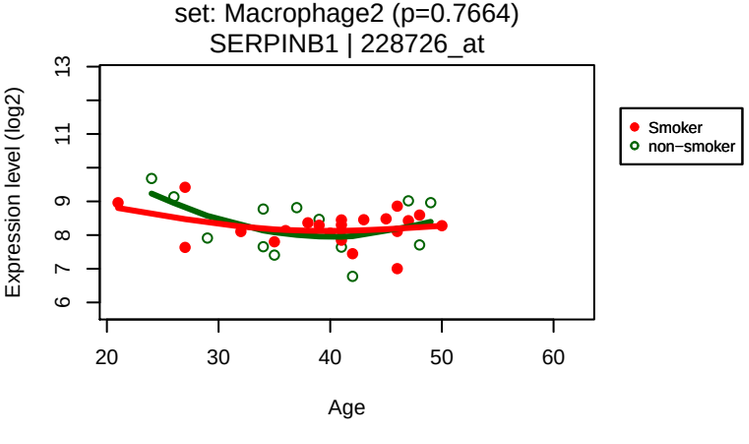

set: SmallAirway2 (p=0.8103)
SERPINB1 | 228726_at
Expression level (log2)
Age
Smoker
non−smoker

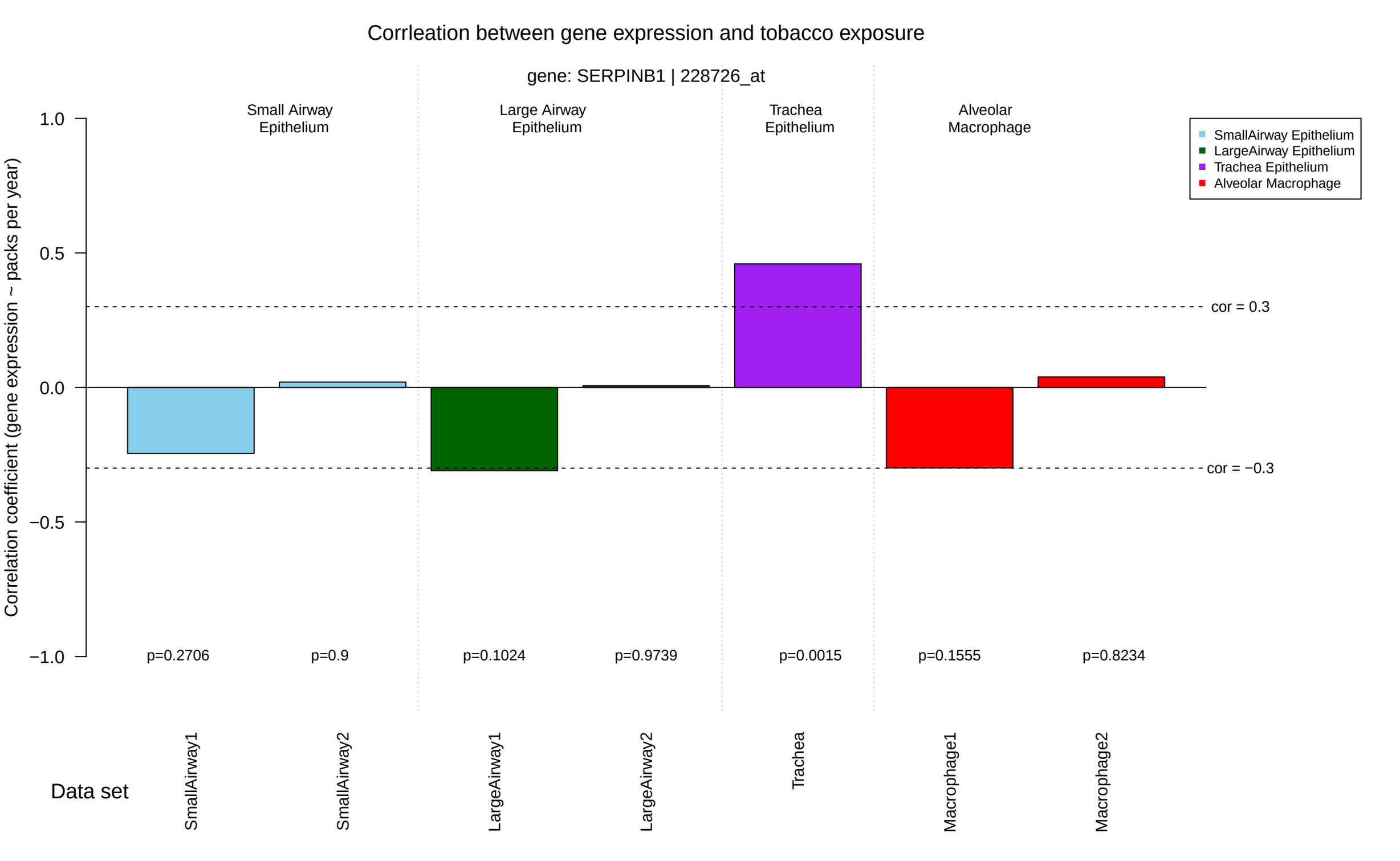

set: LargeAirway1 (p=0.0065)
SERPINB1 | 228726_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6424)
SERPINB1 | 228726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0012)
SERPINB1 | 228726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0416)
SERPINB1 | 228726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7664)
SERPINB1 | 228726_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINB1 | 228726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2706
p=0.9
p=0.1024
p=0.9739
p=0.0015
p=0.1555
p=0.8234
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2